Multi-Core SIV

A sigmoid implied-variance base with zero-wing hats for WW and event smiles

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Abstract

This note documents the Vol-Fitter's sigmoid family, the *Multi-Core SIV* (MC-SIV) model. The base is a six-parameter Sigmoid Implied Variance slice: a log-cosh convexity primitive giving an ATM level, an ATM skew, a convexity amplitude, and *asymmetric* put/call wing steepnesses, C^2 across its centre. On top of the base the model adds R signed *zero-wing hat* kernels — normalized centred second differences of the log-cosh primitive — each of which inserts a local variance hump ($\alpha > 0$) or notch ($\alpha < 0$) *without moving the base wing slopes.* This is exactly the structure a single hyperbola (SVI, Note 02) cannot provide: we prove that a globally convex slice cannot reproduce a WW-shaped smile (central trough with two shoulders), and show that two hats reduce the fit error on such a smile from 153.32 to 8.05 volatility basis points. The model carries $6 + 4R$ parameters, has a closed-form Durrleman butterfly diagnostic, and is calibrated in three stages (base → greedy hat seeding → joint bounded refine) with an analytic Jacobian measured 2.83× faster than finite differences. The cores count R is the user's "cores" slider, the direct analogue of the LQD Legendre order; a note on its tendency to over-fit closes the discussion.

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1 Motivation: what a hyperbola cannot do

SVI's strength — one convex hyperbola — is also its ceiling. Some smiles are not convex. Ahead of a binary event, or in names with bimodal positioning, the implied-volatility curve develops a *WW* shape: a trough near the money, a *shoulder* (local maximum) on each side, and then wings that turn up again toward the Lee asymptotes. Figure 1 shows such a target; the base sigmoid alone (the dotted curve) misses the shoulders by 153.32 vol bps.

Proposition 1 (A convex slice cannot make a WW)**.** *If total variance $w(k)$ is convex on an interval, then on that interval w has no interior strict local maximum, hence no shoulder flanked by two troughs. A WW smile, which has interior local maxima, therefore cannot be represented by any globally convex slice (in particular by raw SVI, whose $w'' > 0$ everywhere).*

Proof. A convex function on an interval lies below the chord between any two points and its one-sided derivatives are non-decreasing; an interior strict local maximum would force the derivative to change from positive to negative, contradicting monotone non-decreasing slope. SVI has $w''(k) = b\sigma^2/r^3 > 0$ (Note 02), so it is strictly convex. □

The MC-SIV answer is to keep a convex *base* that owns the wings and the overall skew, and to add *local, wing-neutral* corrections that can be positive or negative. The design constraint is sharp:

Heuristic

A shoulder is a *local* feature: it changes the body of the smile in a bounded region but must not alter the asymptotic wings (those are pinned by Lee and by liquid risk-reversals). So the correction kernels must vanish — in value, slope, *and* curvature — in both tails. That "zero-wing" requirement is what singles out the centred-second-difference hat of section 3, and it is why an arbitrary Gaussian bump would be wrong (a Gaussian decays but tilts the local skew on the way out).

2 The one-core SIV base

2.1 Coordinates and the log-cosh primitive

Work in the dimensionless log-strike

$$z = \frac{k}{\sigma_{\text{ref}}\sqrt{t}}, \tag{1}$$

where σ_{ref} is the near-the-money quoted volatility, so that z measures moneyness in standard deviations and the model is roughly scale-free across expiries. The convexity primitive is the *log-cosh*

$$\Phi_\kappa(u) = \frac{4}{\kappa^2}\log\cosh\Big(\frac{\kappa u}{2}\Big), \qquad \Phi'_\kappa(u) = \frac{2}{\kappa}\tanh\Big(\frac{\kappa u}{2}\Big), \qquad \Phi''_\kappa(u) = \operatorname{sech}^2\Big(\frac{\kappa u}{2}\Big). \tag{2}$$

Φ_κ is a smooth, convex, even "softened $|u|$": $\Phi_\kappa(u) \to |u|$ for large $|u|$ (the wing) and $\Phi_\kappa(u) \approx u^2/2$ near 0 (the rounded vertex), with κ the transition steepness. For numerical safety the log-cosh is replaced by its asymptote $|x| - \log 2$ beyond $|x| = 50$ to avoid cosh overflow.

2.2 The six base parameters

The base slice and its z-derivatives are

$$
\begin{aligned}
&v_{\text{base}}(z) = V_0 + S_0(z - z_0) + K_0\,\Phi_{\kappa_i}(z - z_0), \\
&v'_{\text{base}}(z) = S_0 + K_0\,\Phi'_{\kappa_i}(z - z_0), \qquad v''_{\text{base}}(z) = K_0\,\Phi''_{\kappa_i}(z - z_0),
\end{aligned}
\tag{3}
$$

with the steepness switching $\kappa_i = \kappa_P$ for $z < z_0$ and $\kappa_i = \kappa_C$ for $z \geq z_0$. The six parameters are the variance level V_0 at the centre z_0, the skew S_0, the convexity amplitude $K_0 \geq 0$, the centre z_0, and the *asymmetric* put/call wing steepnesses κ_P, κ_C. Because Φ and Φ' both vanish at the origin, the slice is C^2 across z_0 even though the steepness switches there — so the asymmetry is in the *rate* at which each wing straightens, not a kink. The asymptotic z-space variance wing slopes are

$$
v'_{\text{base}}(z) \to S_0 \mp \frac{2K_0}{\kappa_{P,C}} \quad (z \to \mp\infty), \tag{4}
$$

i.e. a left slope $S_0 - 2K_0/\kappa_P$ and a right slope $S_0 + 2K_0/\kappa_C$. These are the only handles that set the wings, a fact section 3 exploits.

3 Zero-wing hats

3.1 Definition

The correction kernel is the normalized centred second difference of the log-cosh primitive,

$$
B_{c,h,\kappa}(z) = \frac{\Phi_\kappa(u-h) - 2\Phi_\kappa(u) + \Phi_\kappa(u+h)}{2\Phi_\kappa(h)}, \qquad u = z - c, \tag{5}
$$

centred at c, of half-width h and steepness κ, normalized so that $B_{c,h,\kappa}(c) = 1$. Its derivatives are the same second difference of Φ' and Φ'' over the same normalizer.

Lemma 1 (Zero wings). *$B_{c,h,\kappa}(z) \to 0$, $B'_{c,h,\kappa}(z) \to 0$ and $B''_{c,h,\kappa}(z) \to 0$ as $z \to \pm\infty$.*

Proof. For large $|u|$, $\Phi_\kappa(u) = |u| - \frac{2}{\kappa}\log 2 + o(1)$ is asymptotically *affine*, so the centred second difference $\Phi_\kappa(u-h) - 2\Phi_\kappa(u) + \Phi_\kappa(u+h) \to 0$ (a second difference annihilates affine functions). The same holds for Φ' (which tends to the constants $\pm 2/\kappa$) and Φ'' (which decays). Hence B and its first two derivatives vanish in both tails. □

Lemma 1 is the whole point: adding αB to the base changes the body but leaves the wing slopes (4) *exactly* unchanged. The amplitude α is in variance units and, since $B(c) = 1$, is approximately the variance displacement at the centre. A positive α raises a shoulder; a negative α digs a notch. The curvature at the centre is $B''(c) = 2(\operatorname{sech}^2(\kappa h/2) - 1)/(2\Phi_\kappa(h)) < 0$, so a positive hat is locally concave (a hump), as it should be.

3.2 The full model

The Multi-Core SIV slice is the base plus R signed hats,

$$
\boxed{v_R(z) = v_{\text{base}}(z) + \sum_{r=1}^{R} \alpha_r\, B_{c_r,h_r,\kappa_r}(z),} \qquad w(k) = t\, v_R(z), \tag{6}
$$

with $6 + 4R$ parameters. By lemma 1 every member of this family has the *same* Lee-compatible wings as its base, regardless of R — so adding expressiveness in the body never threatens the tails. The cores count R is the "cores" slider, the analogue of the LQD Legendre order N and the SVI vs LQD vs LV expressiveness dial.

4 Static no-arbitrage

Butterfly-freedom is the Durrleman condition (7), evaluated by converting the z-space variance and its derivatives to total-variance k-space derivatives via $k = \sigma_{\mathrm{ref}}\sqrt{t}\, z$ and $w = tv$:

$$g(k) = \Big(1 - \frac{k\,w'}{2w}\Big)^2 - \frac{w'^2}{4}\Big(\frac{1}{w} + \frac{1}{4}\Big) + \frac{w''}{2} \;\geq 0. \tag{7}$$

Since the risk-neutral density is proportional to g, the diagnostic $g(k) \geq 0$ on the grid is the no-butterfly check. Unlike LQD, MC-SIV is *not* arbitrage-free by construction: a large or narrow hat can drive g negative, so g is reported per fit and the calibration bounds (section 5) keep hats in a benign range. The wings are Lee-compatible by construction (lemma 1); the cross-model wing treatment is Note 09.

5 Calibration

The fit is a three-stage workflow that respects the model's layered structure.

1. **Base fit ($R = 0$).** Fit the six base parameters to the mid quotes under bound constraints, from a data-driven start that reads the ATM variance, the local skew and the local convexity off three interpolated points. This stage is always mid, so its residual is meaningful for the next stage.

2. **Greedy hat seeding.** Compute the base residual in variance space and place each of the R hats at the largest remaining |residual|, masking a neighbourhood after each placement so the hats do not pile up. Each seed starts at half-width $h = 0.40$ and steepness $\kappa = 5.0$ with amplitude clipped to the residual.

3. **Joint bounded refine.** Refine all $6+4R$ parameters together under the requested objective (mid or band) with a mild ridge $\sqrt{\varrho}\,\alpha_r$ on the hat amplitudes, $\varrho = 0.01$. The ridge keeps overlapping cores from exploding against each other without biasing a well-determined amplitude.

All positive parameters (K_0, the steepnesses, the hat half-widths and steepnesses) are bound-constrained directly through scipy's trust-region reflective solver; the hat half-width is held in $[0.15, 1.5]$ and the steepness in $[1.0, 12.0]$. Crucially the cores count is capped so the model never has more parameters than quotes,

$$R \leq \Big\lfloor \frac{N_{\mathrm{quotes}} - 6}{4} \Big\rfloor, \tag{8}$$

which guards sparse short-dated chains against fitting spurious narrow kernels. The refine stage carries the same optional blocks as the other models — band fit (Note 07), var-swap target (Note 08), the model-agnostic calendar hinge (Note 10), and prior persistence (Note 13) — so the sigmoid overlay is no longer a prior exception.

Performance

The dominant cost of an R-core fit is the $(6 + 4R + 1)$-evaluation finite-difference Jacobian. The closed-form Jacobian of the residual stack (data + ridge + calendar) propagated through the log-cosh primitive replaces it in one pass; a fresh run on the two-core WW fit measured 56.6 ms vs 160.1 ms (2.83×) with the optimum unchanged to $1.0e - 16$ in cost. It is gated to the var-swap/prior-free configuration, as in LQD and SVI. See appendix B.

6 Worked example: a WW smile

The target in fig. 1 is a synthetic WW: rising V wings with a shoulder on each side of the ATM trough — the shape proposition 1 forbids a hyperbola. The base SIV ($R = 0$) fits it only to 153.32 vol bps, missing both shoulders. Two hats ($R = 2$, $6 + 4 \cdot 2 = 14$ parameters) reduce the maximum error to 8.05 vol bps while preserving the wing slopes $(S_0 - 2K_0/\kappa_P,\ S_0 + 2K_0/\kappa_C) = (-0.014,\ 0.014)$ and keeping the fit butterfly-free (section 6, $g(k) \geq 0$). Figure 2 shows the decomposition: the convex base owns the V, and the two signed hats add precisely the two shoulders, each localized and wing-neutral.

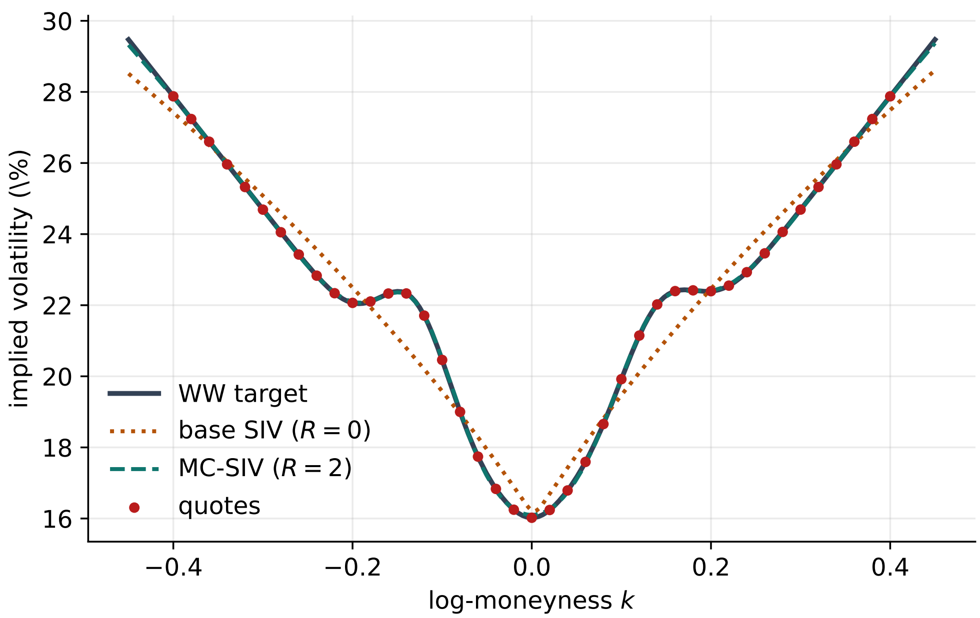
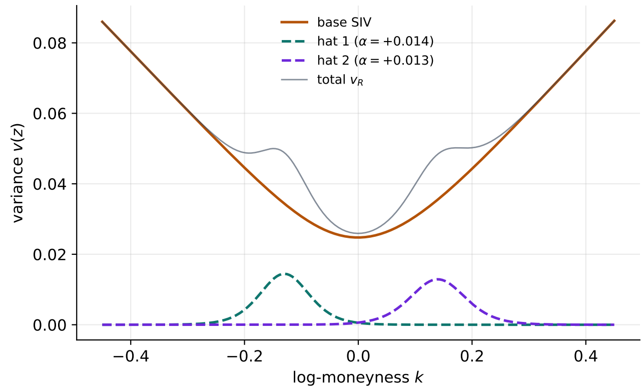
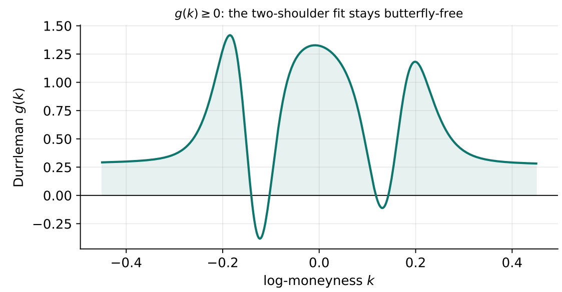


Figure 1: The WW smile: the convex base ($R = 0$, dotted) misses the shoulders by 153.32 vol bps; the two-core MC-SIV fit (dashed) reaches 8.05 vol bps.



(a) base + two signed hats



(b) Durrleman $g(k) \geq 0$

Figure 2: Left: the additive decomposition — the convex base owns the wings, the hats own the shoulders. Right: the butterfly diagnostic stays non-negative.

7 Original ideas, and the over-fit caution

The genuinely original elements are the *zero-wing hat* (5)–lemma 1 — a correction basis that is local in the body and provably neutral in the wings, so expressiveness and tail control are decoupled — and the *layered three-stage calibration* that places the hats on residuals rather than optimizing them blind. Together they let one model span from a clean index smile ($R = 0$, equivalent to a sigmoid SVI) to a bimodal event smile ($R = 2, 3$) on a single slider.

Caution

With great expressiveness comes over-fitting. The offline backtest (project `backtest/`) found that on liquid index chains the Multi-Core hats *over-fit*: cores chase quote noise and manufacture butterfly-arbitrage (measured 60–75% of calibrated nodes with $g < 0$ at $R \geq 1$ on dense data), while the base $R = 0$ behaves like SVI. The identifiability cap (8), the amplitude ridge, and the g diagnostic mitigate this, but the operational guidance is: *use cores only when the smile is genuinely non-convex* (events, bimodal positioning), and prefer LQD or Local-Vol for routine fitting. The cores slider is a scalpel, not a default.

A Hyperparameter atlas

Table 1: Multi-Core SIV hyperparameters.

Knob	Default	Role
Surfaced (FitSettings)		

Knob	Default	Role
`nCores` R	2	Number of zero-wing hats; capped by (8). $0 =$ pure sigmoid base.
`sigmoidRidge` ϱ	0.01	ℓ^2 penalty on hat amplitudes in the refine stage.
`midAnchorWeight`	0.05	Band-fit mid anchor (Note 07).
`weightScheme`	`equal`	Per-quote weights (Note 07).
`calendarWeight`	10^6	Calendar hinge penalty (Note 10).
Internal (seeding / bounds)		
`_H_INIT`	0.40	Hat half-width seed.
`_KAPPA_INIT`	5.0	Hat steepness seed.
`_H_BOUNDS`	$[0.15, 1.5]$	Hat half-width bounds.
`_KAPPA_BOUNDS`	$[1.0, 12.0]$	Hat steepness bounds.
`_C_PAD`	0.5	Centre padding beyond the quoted z-range.
`_V_FLOOR`	10^{-8}	Variance floor so $\sqrt{v}$ stays real.
`_LOGCOSH_CLIP`	50	$\lvert x\rvert$ beyond which $\log\cosh$ uses its asymptote.
`xtol/ftol`	10^{-12}	trf tolerances.

B Performance notes

1. **Analytic Jacobian** through the log-cosh primitive replaces the $(6 + 4R + 1)$-evaluation finite difference; measured $2.83\times$ on the WW fit, optimum unchanged to $1.0e - 16$. Gated to the var-swap/prior-free path.

2. **Layered warm start.** Fitting the base first and seeding hats on its residual gives the joint refine an excellent start, so the bounded trf converges quickly even with several cores.

3. **Vectorized kernels.** Φ, Φ', Φ'' and the hat second differences are NumPy-vectorized and side-effect free, shared between the model and its Jacobian.

4. **Identifiability cap.** Limiting R by the quote count avoids wasting iterations on under-determined narrow kernels and is the first line of defence against the over-fit of section 7.

References

[1] J. Gatheral. *The Volatility Surface: A Practitioner's Guide*. Wiley, 2006.

[2] V. Durrleman. From implied to spot volatilities. *Finance and Stochastics*, 14(2):157–177, 2010.

[3] R. W. Lee. The moment formula for implied volatility at extreme strikes. *Mathematical Finance*, 14(3):469–480, 2004.